MILITARY SPECIFICATION

PRIMER MATERIAL

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for two types of primers.

1.2 Classification. The primers shall be of the following types:

Type I	Metal primer
Type II	Rubber primer

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following document of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by contractors in connection with procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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2.2 Non-Government documents. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

STANDARDS

American Society for Testing and Materials (ASTM)

ASTM-D-1475	Density of Paint, Varnish Lacquer, and Related Products.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS.

3.1 Product characteristics. Type I primer shall be a gray opaque liquid. Type II primer shall be a black opaque liquid.

3.2 Chemical and physical properties. Chemical and physical properties of the primers shall conform to the requirements of Table I.

TABLE I. Chemical and physical properties.

Property	Type I		Type II	
	Min	Max	Min	Max
Solids, percent	22.0	26.0	23.0	26.5
Density at 25°C, lb/gal	7.6	8.1	8.9	9.2
Viscosity at 25°C, centipoise	85	165	450	900

3.3 Stability. When stored at temperatures of 5 to 32 degrees Celsius (°C), the primers shall meet all the requirements of this specification for a minimum of 12 months from receipt. The shelf life for both types of primer may be extended for 6-month intervals by reacceptance testing for conformance to Table I requirements.

3.4 Toxic products and safety. Safety regulations and guidelines applicable to the use of the primer should be complied with to preclude personal injury and damage to equipment and facilities.

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ds shall be formed and screws used for electrical connections shall be groups running from hinged pane

3.5 Workmanship. The workmanship shall be such that the primers are uniform, of consistent high quality and free from visible contamination.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor shall be responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. The lot shall be sampled in accordance with Table II.

TABLE II. Sampling plan.

Number of containers in lot	Number of containers sampled (primary sample)	Number of composite samples
100 or more	10% (nearest whole number)	5
51 - 99	10	4
11 - 50	10	3
1 - 10	All	2

4.2.1 Primary samples. Physical properties tests shall be run on each primary sample (see Table II). The material container may be sampled by use of a clean glass tube, rod, or pipet. If the container is small enough to be handled safely, the sample may be obtained by pouring. The smallest sample size possible that is consistent with test requirements shall be taken. The minimum sample size shall be two ounces. Glass containers shall be used for all liquid samples. Each sample shall be labeled with date, lot number, and manufacturer's container identification number. Failure of any primary sample to pass all of the physical properties tests herein shall result in rejection of the lot represented.

4.2.2 Composite samples. Chemical properties tests shall be run on each composite sample. Divide the primary samples equally into the number of composites shown in Table II. Blend each composite thoroughly by manipulation of the container. Label each composite with Roman numerals,

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also include date, lot numbers, and manufacturer's container identification numbers. The remainder of the primary samples shall be retained pending acceptance or rejection of the lot. Failure of any composite sample to pass all of the chemical properties tests herein shall result in rejection of the lot represented.

4.3 Quality conformance inspections and tests. Quality conformance inspections and tests shall consist of the following:

a. Tests of Table I properties (see 4.4).

b. Inspection of filled containers (see 4.5.1).

c. Visual inspection (see 4.5.2).

4.4 Test methods. Tests shall be performed using apparatus, reagents, and procedures specified herein. The use of alternate apparatus, reagents, or procedures shall require prior written approval of the procuring activity.

4.4.1 Solids. Weigh 0.8 to 1.2 grams (gm) of sample to the nearest 0.1 milligram into a tared aluminum disposable weighing dish. Dry for 2 hours at 68 ± 1°C. Cool sample in a desiccator and reweigh. Calculate solids as follows:

$$\text{Solids, percent} = \frac{100(C - A)}{B - A}$$

Where

A = weight of dish, gm

B = weight of dish and sample, gm

C = weight of dish and sample, after heating, gm

Report solids to the nearest 0.1 percent.

4.4.2 Density at 25°C. Determine the weight per gallon in accordance with ASTM-D-1475. Report the weight per gallon to the nearest 0.1 pound per gallon.

4.4.3 Viscosity at 25°C. Determine viscosity at 25°C with a Brookfield Model RVT Viscometer using spindle No. 1 at 20 rpm (Type I primer) and spindle No. 6 at 5 rpm (Type II primer). Take a minimum of five readings and average the readings. Calculate and report viscosity to the nearest centipoise.

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4.5 Examinations.

4.5.1 Inspection of filled containers. All filled containers shall be inspected prior to shipment or use for accuracy of markings and for defects in containers and closures. All defective containers and closures shall be repaired or replaced, and contents therein shall be re-inspected prior to shipment or use.

4.5.2 Visual inspection. All samples shall be visually inspected to determine conformance to the requirements of 3.1 and 3.5.

4.6 Records. Certification and test data shall be prepared as required by the procuring activity (see 6.2.2).

5. PACKAGING.

5.1 Packaging and packing. Unless otherwise specified in the contract or purchase order (see 6.2.1), packaging and packing of the primers shall be in accordance with commercial practice to ensure carrier acceptance and shall be of such construction and materials that the contents will be adequately protected against loss or contamination.

5.2 Marking for shipment. Unless otherwise specified in the contract or purchase order (see 6.2.1), each shipping container shall be marked in accordance with the requirements of MIL-STD-129. Container marking shall include the following:

a. The supplier's lot number

b. Procuring activity purchase order number

c. Container identification number (applied in numerical sequence as the containers are filled)

d. Date of manufacture

e. Manufacturer's Code Ident

f. Net and tare weight of the container

g. Material identification

6. NOTES AND CONCLUDING MATERIAL.

6.1 Intended use. The intended use of the material described herein is as a bond promoter in bonding rubber to metal and to other elastomers.

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6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Responsibility for inspection and inspection facilities if different than 4.1.

c. Special packaging, packing, or shipping requirements, if applicable (see Section 5).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.6	Certification	UDI-A-23264B
	Test Data	DI-T-4024

(Copies of DIDs required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Lot. At place of manufacture, a lot consists of one batch (see 6.3.2) or a uniform blend of two or more batches. At place of delivery, a lot consists of primer from one supplier's lot received in a single shipment. Partial shipments may be considered as a single shipment by the procuring activity.

6.3.2 Batch. A batch consists of primer made as one unit in an unchanged manufacturing process.

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6.4 Suggested source of supply. Products that have met the requirements of this specification in past procurement actions are marketed by Hughson Chemicals, Lord Corporation, Code Ident 30676 as Chemlok 205 (Type I) and Chemlok 234B (Type II). This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification nor as any limitation of additional potential sources of supply.

Preparing activity:
Navy - AS

(Project 6810-NB28)

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DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-P-85499(AS), PRIMER MATERIAL

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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DD FORM 1426 1 OCT 76

Replaces edition of 1 Jan 72 which may be used.

S/N 0102-LF-001-4260

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